2011 F=ma Exam: Problem 17

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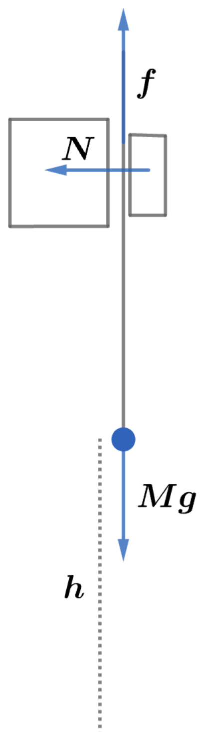

From the previous problem, Jonathan applies a normal force $N = Mg/\mu_s$. After Becky makes a small jump, friction becomes kinetic since the rope starts slipping. The energy dissipated as Becky slides down is

$$E_d = fh = \mu_k N h = \frac{\mu_k M g h}{\mu_s}$$

Becky's initial gravitational potential energy is converted to kinetic energy as she falls. Conserving energy, we have

$$E_i - E_d = E_f$$
$$Mgh - \frac{\mu_k M g h}{\mu_s} = \frac{1}{2} M v^2$$
$$\frac{1}{2} v^2 = gh\left(1 - \frac{\mu_k}{\mu_s}\right)$$
$$v = \sqrt{2gh\left(1 - \frac{\mu_k}{\mu_s}\right)}$$

so the answer is $\boxed{\text{D}}$.